Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	14 July 2026
Team ID	________________
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technical Architecture

User → Web Application → Data Processing Module → MySQL Database → Analytics Engine → Dashboard & Reports

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web application	HTML, CSS, JavaScript
2	Application Logic-1	Business logic	Python (Flask)
3	Application Logic-2	Data analysis	Pandas, NumPy
4	Application Logic-3	Visualization	Matplotlib
5	Database	Electricity data	MySQL
6	Cloud Database	Cloud storage	Firebase / Google Cloud SQL
7	File Storage	Dataset files	Local Filesystem
8	External API-1	Weather (optional)	OpenWeather API
9	External API-2	Power statistics	Government Open Data API
10	Machine Learning Model	Consumption prediction	Linear Regression
11	Infrastructure	Deployment	Local / Google Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Web framework	Flask
2	Security	Authentication & encryption	SHA-256, HTTPS
3	Scalable Architecture	Three-tier architecture	Flask + MySQL
4	Availability	Reliable deployment	Cloud Hosting
5	Performance	Fast querying & charts	Indexed MySQL